

EC400: Optimization and Fixed Points

Lecture 2: Static Optimization

Francesco Nava

London School of Economics

A Pre-Sessional Course

Optimization:

Unconstrained Optimization

Extreme Points

Definition Extreme Points

Definition

Let $f(\mathbf{x})$ be a real valued function defined on a subset C of $\mathbb{R}^n$. Then, $\mathbf{x}^* \in C$ is:

- a **global maximizer** for $f(\mathbf{x})$ on C if

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) \text{ for all } \mathbf{x} \in C$$

- a **strict global maximizer** for $f(\mathbf{x})$ on C if

$$f(\mathbf{x}^*) > f(\mathbf{x}) \text{ for all } \mathbf{x} \in C \text{ such that } \mathbf{x} \neq \mathbf{x}^*$$

Definition

The ball $B(\mathbf{x}, \varepsilon)$ centred at $\mathbf{x}$ of radius ε is the set of all vectors $\mathbf{y}$ in $\mathbb{R}^n$ whose distance from $\mathbf{x}$ is less than ε :

$$B(\mathbf{x}, \varepsilon) = \{\mathbf{y} \in \mathbb{R}^n \mid \varepsilon > \|\mathbf{y} - \mathbf{x}\|\}$$

Definition Extreme Points

Definition

Let $f(\mathbf{x})$ be a real valued function defined on a subset C of $\mathbb{R}^n$. Then, $\mathbf{x}^* \in C$ is:

- a **local maximizer** for $f(\mathbf{x})$ if $\varepsilon > 0$ exists such that:

$$f(\mathbf{x}^*) \geq f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in C \cap B(\mathbf{x}^*, \varepsilon)$$

- a **strict local maximizer** for $f(\mathbf{x})$ if $\varepsilon > 0$ exists such that:

$$f(\mathbf{x}^*) > f(\mathbf{x}) \quad \text{for all } \mathbf{x} \in C \cap B(\mathbf{x}^*, \varepsilon) \text{ such that } \mathbf{x} \neq \mathbf{x}^*$$

- a **critical point** for $f(\mathbf{x})$ if first partial derivatives exist at $\mathbf{x}^*$ and if:

$$\frac{\partial f(\mathbf{x}^*)}{\partial x_i} = 0 \quad \text{for } i = 1, 2, \dots, n \quad \Leftrightarrow \quad Df(\mathbf{x}^*) = \mathbf{0}$$

Do Extreme Points Exist?

Definition

A set is **compact** if it is closed and bounded.

Theorem (Extreme Value Theorem)

Let C be a compact subset of $\mathbb{R}^n$ and $f(x)$ be a continuous function defined on C . If so, there exists:

- (1) a point $\mathbf{x}^* \in C$ which is a global maximizer of f ;*
- (2) a point $\mathbf{x}_* \in C$ which is a global minimizer of f .*

That is,

$$f(\mathbf{x}_*) \leq f(\mathbf{x}) \leq f(\mathbf{x}^*) \text{ for all } x \in C.$$

Necessary and Sufficient Conditions

Univariate Functions: FOC and SOC

Let $f(x)$ be a continuously differentiable function on an interval I .

First Order Necessary Conditions for a maximum in $\mathbb{R}$.

If x^* is a **local maximizer** of $f(x)$ then:

- either x^* is an end point of I , or $f'(x^*) = 0$.

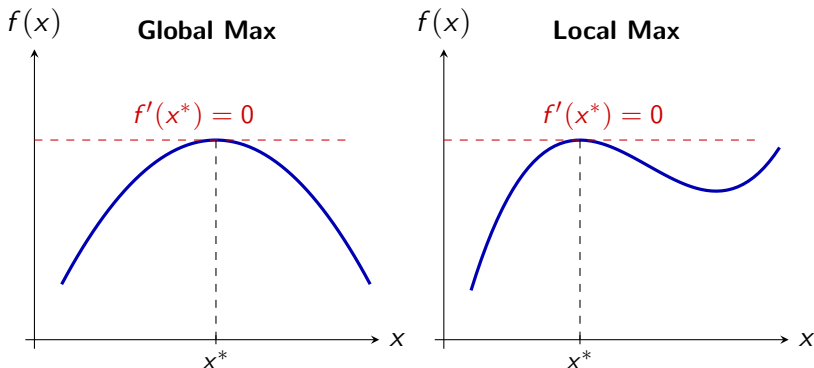
Second Order Sufficient Conditions for a maximum in $\mathbb{R}$.

If $f(x)$ is twice continuously differentiable on I and $f'(x^*) = 0$ then x^* is:

- a **global maximizer** of f on I , if $f''(x) \leq 0$ for all $x \in I$.
- a **strict global maximizer** of f on I , if $f''(x) < 0$ for all $x \in I$.
- a **strict local maximizer** of f on I , if $f''(x^*) < 0$.

Univariate Functions: FOC and SOC

Global and local maxima display by flatness and local concavity:



Multivariate Functions: FOC and Local SOC

Let $f(\mathbf{x})$ be a twice continuously differentiable function on $C \subset \mathbb{R}^n$.

First Order Necessary Conditions for a maximum in $\mathbb{R}^n$.

If $\mathbf{x}^*$ is an interior point of C and:

- a **local extremum** of f , then $Df(\mathbf{x}^*) = \mathbf{0}$.
- a **local maximum** of f , then $D^2f(\mathbf{x}^*)$ is **negative semidefinite**.
- a **local minimum** of f , then $D^2f(\mathbf{x}^*)$ is **positive semidefinite**.

Second Order Sufficient Conditions for a local maximum in $\mathbb{R}^n$.

If $\mathbf{x}^*$ is critical point of f , then $\mathbf{x}^*$ is:

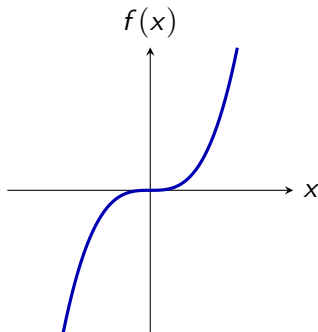
- a **strict local maximizer** of f , if $D^2f(\mathbf{x}^*)$ is **negative definite**;
- a **strict local minimizer** of f , if $D^2f(\mathbf{x}^*)$ is **positive definite**.

Multivariate Functions: More on SOC

But, it is **not** true that if $\mathbf{x}^*$ is a critical point and $D^2f(\mathbf{x}^*)$ is negative (positive) **semidefinite**, then $\mathbf{x}^*$ is a local maximum (minimum).

A counterexample is $f(x) = x^3$ which has a critical point at $x^* = 0$:

- at which the Hessian is semidefinite, $D^2f(0) = 0$;
- but which is neither a maximum nor minimum; it's a saddle point.



Multivariate Functions: Local SOC Intuition

To determine **whether a critical point is a local extreme point** we consider the **Hessian** of the map f (the matrix of the second derivatives).

This is the case as the Hessian captures the change in slope.

At a local maximum, the function:

- has to be flat and
- its slope has to decline.

Similar intuition applies to global SOC.

The Hessian is always a **symmetric matrix**, since cross-partial derivatives are equal by the Clairaut-Schwarz Theorem.

Multivariate Functions: Local SOC Proof

Consider the Taylor Expansion:

$$f(\mathbf{x}^* + \mathbf{h}) = f(\mathbf{x}^*) + Df(\mathbf{x}^*)\mathbf{h} + \frac{1}{2}\mathbf{h}^T D^2f(\mathbf{x}^*)\mathbf{h} + R_2(\mathbf{h}|\mathbf{x}^*).$$

Ignore $R_2(\mathbf{h}|\mathbf{x}^*)$, set $Df(\mathbf{x}^*) = 0$, and observe that:

$$f(\mathbf{x}^* + \mathbf{h}) - f(\mathbf{x}^*) \approx \frac{1}{2}\mathbf{h}^T D^2F(\mathbf{x}^*)\mathbf{h}.$$

If $D^2f(\mathbf{x}^*)$ is negative definite, then the right hand side is negative.

This in turn implies that for small enough $\mathbf{h}$:

$$f(\mathbf{x}^* + \mathbf{h}) < f(\mathbf{x}^*).$$

Hence, $\mathbf{x}^*$ is a strict local maximizer of f .

Example: Unconstrained Optimization

Consider the function:

$$f(\mathbf{x}) = x_1^3 - x_2^3 + 9x_1x_2.$$

Find critical points, $(0, 0)$ and $(3, -3)$, setting the gradient to zero:

$$Df(\mathbf{x}) = (3x_1^2 + 9x_2, -3x_2^2 + 9x_1) = \mathbf{0}.$$

The Hessian satisfies:

$$D^2f(\mathbf{x}) = \begin{pmatrix} 6x_1 & 9 \\ 9 & -6x_2 \end{pmatrix}.$$

The first order LPM is $6x_1$ and the second order LPM is $-36x_1x_2 - 81$.

- $(0, 0)$ is a **saddle point** since the Hessian is indefinite.
- $(3, -3)$ is a **strict local minimum** since the Hessian is positive definite.
- However, $(3, -3)$ **is not a global minimum** (consider $\mathbf{x} \rightarrow (0, \infty)$).

Global SOC and Concavity

Multivariate Functions: Global SOC

Let $f(\mathbf{x})$ be a twice continuously differentiable function on $\mathbb{R}^n$.

Second order sufficient conditions for global extrema in $\mathbb{R}^n$.

Suppose that $\mathbf{x}^*$ is a critical point of f , then $\mathbf{x}^*$ is:

- a **global maximizer**, if $D^2f(\mathbf{x})$ is **negative semidefinite** on $\mathbb{R}^n$.
- a **global minimizer**, if $D^2f(\mathbf{x})$ is **positive semidefinite** on $\mathbb{R}^n$.
- a **strict global maximizer**, if $D^2f(\mathbf{x})$ is **negative definite** on $\mathbb{R}^n$.
- a **strict global minimizer**, if $D^2f(\mathbf{x})$ is **positive definite** on $\mathbb{R}^n$.

Remark

If f is a differentiable, concave function defined on a convex set U of $\mathbb{R}^n$ and $\mathbf{x}, \mathbf{y} \in U$, then:

$$Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq 0 \quad \Rightarrow \quad f(\mathbf{y}) \leq f(\mathbf{x}).$$

Thus, $\mathbf{x}^ \in U$ is a global maximizer of f , if one of the following holds:*

- *$Df(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \leq 0$ for all $\mathbf{y} \in U$;*
- *$Df(\mathbf{x}^*) = 0$.*

Proof of Remark: Recall that for differentiable functions concavity was equivalent to

$$f(\mathbf{y}) - f(\mathbf{x}) \leq Df(\mathbf{x})(\mathbf{y} - \mathbf{x}).$$

Thus, we immediately obtain that

$$Df(\mathbf{x})(\mathbf{y} - \mathbf{x}) \leq 0 \Rightarrow f(\mathbf{y}) \leq f(\mathbf{x}).$$

But if so, when $Df(\mathbf{x}^*) = \mathbf{0}$ the result obtains as

$$Df(\mathbf{x}^*)(\mathbf{y} - \mathbf{x}^*) \leq 0 \text{ for all } \mathbf{y} \in U.$$

Thus, $\mathbf{x}^*$ maximizes f . $\square$

Why Care about Concavity?

- The property that critical points of concave functions are global maximizers is an important one in economic theory.
- For example, many economic principles, such as marginal rate of substitution equals the price ratio, or marginal revenue equals marginal cost are simply the first order necessary conditions of the corresponding maximization problems (as we will see).
- Ideally, such conditions should also be sufficient to identify optimizing behavior. But this will indeed be the case when the objective functions are concave.

Summary: Unconstrained FOC and SOC

Summary: Unconstrained FOC and SOC

Consider a subset $U \subseteq \mathbb{R}^n$, a point $\mathbf{x}^* \in U$, and a twice continuously differentiable objective function $f : U \rightarrow \mathbb{R}$.

Let $\bar{B}(\mathbf{x}|\varepsilon) = B(\mathbf{x}|\varepsilon) \setminus \{\mathbf{x}\}$.

Consider the following conditions:

(A) $Df(\mathbf{x}^*) = 0$ & $D^2f(\mathbf{x}^*) \leq 0$.

(B) $Df(\mathbf{x}^*) = 0$, $D^2f(\mathbf{x}^*) \leq 0$, &
 $D^2f(\mathbf{x}) < 0$ for any $\mathbf{x} \in \bar{B}(\mathbf{x}^*|\varepsilon)$ for some $\varepsilon > 0$.

(C) $Df(\mathbf{x}^*) = 0$ & $D^2f(\mathbf{x}) \leq 0$ for any $\mathbf{x} \in U$.

(D) $Df(\mathbf{x}^*) = 0$ & $D^2f(\mathbf{x}) \leq 0$ for any $\mathbf{x} \in B(\mathbf{x}^*|\varepsilon)$ for some $\varepsilon > 0$.

(E) $Df(\mathbf{x}^*) = 0$, $D^2f(\mathbf{x}) \leq 0$ for any $\mathbf{x} \in U$ &
 $D^2f(\mathbf{x}) < 0$ for any $\mathbf{x} \in \bar{B}(\mathbf{x}^*|\varepsilon)$ for some $\varepsilon > 0$.

Summary: Unconstrained FOC and SOC

The table summarizes necessary and sufficient conditions for a maximum:

	Necessary	Sufficient
Strict Global Max	A	E
Global Max	A	C
Strict Local Max	A	B
Local Max	A	D

Optimization:

Constrained Optimization

The Constrained Optimization Problem

The Constrained Optimization Problem

Consider the following problem:

$$\begin{aligned} \max_{\mathbf{x}} \quad & f(\mathbf{x}) \quad \text{s.t.} \\ & g_1(\mathbf{x}) \leq 0, \dots, g_k(\mathbf{x}) \leq 0, \\ & h_1(\mathbf{x}) = 0, \dots, h_m(\mathbf{x}) = 0. \end{aligned}$$

The function $f(\mathbf{x})$ is called **the objective function**.

The functions $\mathbf{g}(\mathbf{x})$ and $\mathbf{h}(\mathbf{x})$ are the **constraint functions**:

- k **inequality** constraint $\mathbf{g}(\mathbf{x}) \leq \mathbf{0}$;
- m **equality** constraints $\mathbf{h}(\mathbf{x}) = \mathbf{0}$.

Example: Utility Maximization

Consider the following **utility maximization** problem:

$$\begin{aligned} \max_{x_1, \dots, x_n} \quad & U(x_1, x_2, \dots, x_n) \quad \text{s.t.} \\ & p_1 x_1 + p_2 x_2 + \dots + p_n x_n \leq I, \\ & x_1 \geq 0, \dots, x_n \geq 0. \end{aligned}$$

Note that we can treat the latter constraints as:

$$-x_i \leq 0$$

Equality Constraints

Intuition: 2 Variables and 1 Equality Constraint

A simple case:

$$\begin{aligned} \max_{x_1, x_2 \geq 0} \quad & f(x_1, x_2) \\ \text{s.t.} \quad & p_1 x_1 + p_2 x_2 = I \end{aligned}$$

Geometrical representation:

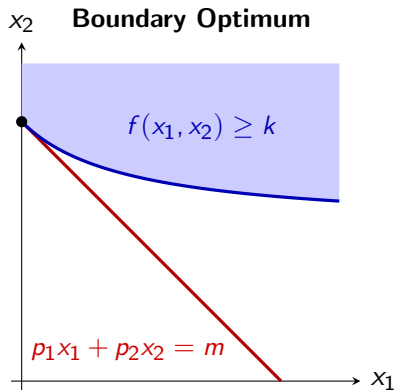
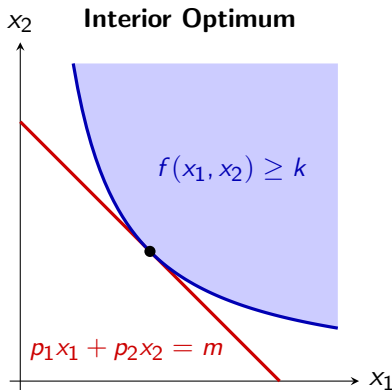
- 1 Draw the constraint on the (x_1, x_2) plane.
- 2 Draw the level curves of the objective function f .
- 3 Find the highest value level curve which meets the constraint.

Such a level curve cannot cross the constraint, therefore it must:

- 1 either be **tangent** to the constraint;
- 2 or be at **a corner**.

Intuition: 2 Variables and 1 Equality Constraint

The maximum must be at a tangency point or at a corner:



Intuition: 2 Variables and 1 Equality Constraint

Omitting non negativity constraints, consider the two-variable problem:

$$\max_{\mathbf{x} \in \mathbb{R}^2} f(\mathbf{x}) \quad \text{s.t.} \quad h(\mathbf{x}) = 0.$$

To find the slope of the level set $f(\mathbf{x}) = a$, use total differentiation,

$$\frac{\partial f(\mathbf{x})}{\partial x_1} dx_1 + \frac{\partial f(\mathbf{x})}{\partial x_2} dx_2 = 0 \implies \frac{dx_2}{dx_1} = -\frac{\partial f(\mathbf{x})}{\partial x_1} / \frac{\partial f(\mathbf{x})}{\partial x_2}.$$

So, the slope of the level curve of f at the solution $\mathbf{x}^*$ equals

$$-\frac{\partial f}{\partial x_1}(\mathbf{x}^*) / \frac{\partial f}{\partial x_2}(\mathbf{x}^*).$$

Similarly, the slope of the constraint at the solution $\mathbf{x}^*$ satisfies

$$-\frac{\partial h}{\partial x_1}(\mathbf{x}^*) / \frac{\partial h}{\partial x_2}(\mathbf{x}^*).$$

Intuition: 2 Variables and 1 Equality Constraint

If the solution $\mathbf{x}^*$ is interior, tangency requires

$$\frac{\partial f}{\partial x_1}(\mathbf{x}^*) / \frac{\partial f}{\partial x_2}(\mathbf{x}^*) = \frac{\partial h}{\partial x_1}(\mathbf{x}^*) / \frac{\partial h}{\partial x_2}(\mathbf{x}^*).$$

If so, let μ be common value satisfying

$$\mu \equiv \frac{\partial f}{\partial x_i}(\mathbf{x}^*) / \frac{\partial h}{\partial x_i}(\mathbf{x}^*) \quad \text{for all } i.$$

An interior solution to this problem is thus pinned down by the system of three equations in three unknowns (x_1, x_2, μ) :

$$\begin{aligned} \frac{\partial f}{\partial x_1}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_1}(\mathbf{x}^*) &= 0, \\ \frac{\partial f}{\partial x_2}(\mathbf{x}^*) - \mu \frac{\partial h}{\partial x_2}(\mathbf{x}^*) &= 0, \\ h(\mathbf{x}^*) &= 0. \end{aligned}$$

Intuition: 2 Variables and 1 Equality Constraint

Equivalently, to find a solution first form the **Lagrangian** function:

$$L(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu h(\mathbf{x}),$$

and then find the **critical point of L** , by setting:

$$\frac{\partial L}{\partial x_1} = 0; \quad \frac{\partial L}{\partial x_2} = 0; \quad \frac{\partial L}{\partial \mu} = 0,$$

as this yields the same system of equations as above.

The variable μ is called the **Lagrange multiplier**.

The previous argument reduces a constrained problem in two variables to an unconstrained problem in three variables.

Lagrange Constraint Qualification

Caveat: For slopes to be defined, we cannot have $D\mathbf{h}(\mathbf{x}^*) = \mathbf{0}$.

Definition (Lagrange Constraint Qualification)

With a single constraint, at the solution $\mathbf{x}^*$ we must have that:

$$D\mathbf{h}(\mathbf{x}^*) \neq \mathbf{0}.$$

Similarly, with multiple constraints, at the solution $\mathbf{x}^*$ we must have that $D\mathbf{h}(\mathbf{x}^*)$ is linearly independent.

Simple Lagrange Theorem

Theorem (Simple Lagrange Theorem)

Let f and h be twice continuously differentiable in $\mathbb{R}^2$. Let $\mathbf{x}^* = (x_1^*, x_2^*)$ be a solution to

$$\max_{x_1, x_2} f(x_1, x_2) \quad \text{s.t.} \quad h(x_1, x_2) = 0,$$

and let $Dh(\mathbf{x}^*) \neq \mathbf{0}$. Then $(\mathbf{x}^*, \mu^*)$ is a *critical point of the Lagrangian* for some real number μ^* .

Intuition:

The Lagrangian penalizes the objective when violating the constraint.

No penalty is paid at the optimum as the constraint holds.

The multiplier is the smallest fine for which the constraint holds.

Example: Equality Constraints

Consider the following example:

$$\begin{array}{ll}\max_{x_1, x_2} & x_1 x_2 \\ \text{s.t.} & x_1 + 4x_2 = 16.\end{array}$$

The constraint qualification is satisfied. The Lagrangian is

$$L(x_1, x_2, \mu) = x_1 x_2 - \mu(x_1 + 4x_2 - 16).$$

While the first order conditions are:

$$\begin{aligned}x_2 - \mu &= 0, \\ x_1 - 4\mu &= 0, \\ x_1 + 4x_2 - 16 &= 0.\end{aligned}$$

The only solution is then $x_1 = 8$, $x_2 = 2$, $\mu = 2$.

General Lagrange Theorem

With many variables, not much changes and we still have

$$Df(\mathbf{x}^*) = \mu Dh(\mathbf{x}^*) \quad \text{and} \quad h(\mathbf{x}^*) = 0.$$

With several constraints instead we have that:

Theorem (General Lagrange Theorem)

Let f and h_i for all $i \in \{1, \dots, m\}$ be continuously differentiable functions from $\mathbb{R}^n$ to $\mathbb{R}$. Let $\mathbf{x}^*$ maximize $f(\mathbf{x})$ subject to $\mathbf{h}(\mathbf{x}) = \mathbf{0}$ and let $D\mathbf{h}(\mathbf{x}^*)$ be linearly independent. If so, there is a vector $\boldsymbol{\mu}^* = (\mu_1^*, \dots, \mu_m^*)$ such that

$$\underbrace{Df(\mathbf{x}^*)}_{1 \times n} = \underbrace{\boldsymbol{\mu}^*}_{1 \times m} \underbrace{D\mathbf{h}(\mathbf{x}^*)}_{m \times n} = \sum_{i=1}^m \mu_i^* Dh_i(\mathbf{x}^*).$$

There are **no restrictions on the sign of μ** .

[Details in Notes]

Extra: Proving Lagrange Theorem

Define the following auxiliary function

$$f^a(\mathbf{x}) = f(\mathbf{x}) - a \|\mathbf{h}(\mathbf{x})\|^2 / 2 - c \|\mathbf{x} - \mathbf{x}^*\|^2 / 2.$$

Let $\mathbf{x}^a$ maximize $f^a(\mathbf{x})$ subject to $\|\mathbf{x} - \mathbf{x}^*\| \leq \varepsilon$. By optimality of $\mathbf{x}^a$,

$$f^a(\mathbf{x}^a) \geq f^a(\mathbf{x}^*) = f(\mathbf{x}^*).$$

As $f^a(\mathbf{x})$ is bounded on $\|\mathbf{x} - \mathbf{x}^*\| \leq \varepsilon$ by continuity, we have that

$$\lim_{a \rightarrow \infty} \|\mathbf{h}(\mathbf{x}^a)\|^2 = 0.$$

Thus, for $\bar{\mathbf{x}} = \lim_{a \rightarrow \infty} \mathbf{x}^a$, taking limits of the optimality condition requires

$$f(\bar{\mathbf{x}}) - c \|\bar{\mathbf{x}} - \mathbf{x}^*\|^2 / 2 = \lim_{a \rightarrow \infty} f^a(\mathbf{x}^a) \geq f(\mathbf{x}^*).$$

As $f(\bar{\mathbf{x}}) \leq f(\mathbf{x}^*)$ by optimality of $\mathbf{x}^*$, we get that $\bar{\mathbf{x}} = \mathbf{x}^*$.

Extra: Proving Lagrange Theorem

Next observe that optimality of $\mathbf{x}^a$ implies

$$Df^a(\mathbf{x}^a) = \underbrace{(Df(\mathbf{x}^a) - c(\mathbf{x}^a - \mathbf{x}^*)^T)}_{1 \times n} - \underbrace{a\mathbf{h}(\mathbf{x}^a)^T}_{1 \times m} \underbrace{D\mathbf{h}(\mathbf{x}^a)}_{m \times n} = 0. \quad (1)$$

By LI, we can post-multiply by $D\mathbf{h}(\mathbf{x}^a)^T [D\mathbf{h}(\mathbf{x}^a)D\mathbf{h}(\mathbf{x}^a)^T]^{-1}$ and get

$$(Df(\mathbf{x}^a) - c(\mathbf{x}^a - \mathbf{x}^*)^T)D\mathbf{h}(\mathbf{x}^a)^T [D\mathbf{h}(\mathbf{x}^a)D\mathbf{h}(\mathbf{x}^a)^T]^{-1} = a\mathbf{h}(\mathbf{x}^a)^T.$$

Taking limits as a diverges, then implies that

$$\lim_{a \rightarrow \infty} a\mathbf{h}(\mathbf{x}^a)^T = Df(\mathbf{x}^*)D\mathbf{h}(\mathbf{x}^*)^T [D\mathbf{h}(\mathbf{x}^*)D\mathbf{h}(\mathbf{x}^*)^T]^{-1} \equiv \boldsymbol{\mu}^*.$$

Taking limits and substituting into (1) then implies the desired result

$$Df(\mathbf{x}^*) = \boldsymbol{\mu}^* D\mathbf{h}(\mathbf{x}^*).$$

Inequality Constraints

Intuition: 2 Variables and 1 Inequality Constraint

Consider the following problem:

$$\begin{array}{ll}\max_{x_1, x_2} & f(x_1, x_2) \\ \text{s.t.} & g(x_1, x_2) \leq 0.\end{array}$$

The geometrical representation implies that the solution can be:

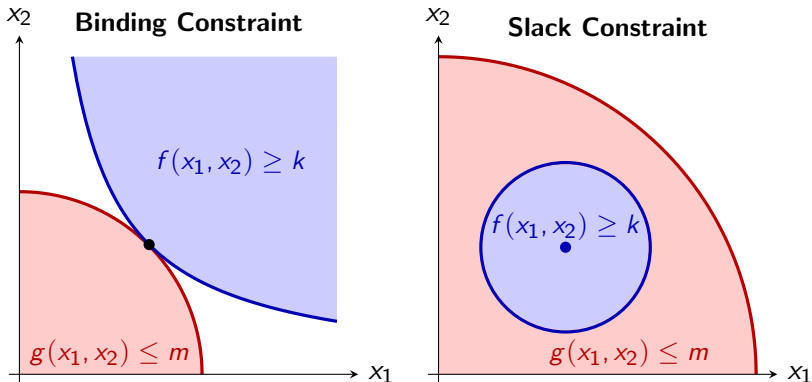
- ① in the **interior** of the constraint set;
- ② on the **boundary** of the the constraint set.

In case 2, again the level curve cannot cross the constraint at the solution; therefore it must be

- either **tangent** to the constraint set;
- or at **a corner**.

Intuition: 2 Variables and 1 Inequality Constraint

The maximum must be interior or on the boundary:



Intuition: 2 Variables and 1 Inequality Constraint

Question: When the constraint binds, is the problem equivalent to a problem with an equality constraint?

Answer: Yes, but we need a restriction on the Lagrange multiplier.

The gradients are again in line so that one is multiple of the other:

$$Df(x^*) = \mu Dg(x^*)$$

However, now the **sign of μ** is now important:

- the gradients must point in the same direction;
- or it is possible to increase f while satisfying the constraint.

The main difference between inequality and equality constraints is that Lagrange multiplier of a binding constraint must satisfy **$\mu \geq 0$** .

Intuition: 2 Variables and 1 Inequality Constraint

Consider the **Lagrangian**:

$$L(x_1, x_2, \mu) = f(x_1, x_2) - \mu g(x_1, x_2)$$

As usual, we find the **critical point of L** , by setting:

$$\begin{aligned}\frac{\partial L}{\partial x_1} &= \frac{\partial f}{\partial x_1} - \mu \frac{\partial g}{\partial x_1} = 0 \\ \frac{\partial L}{\partial x_2} &= \frac{\partial f}{\partial x_2} - \mu \frac{\partial g}{\partial x_2} = 0\end{aligned}$$

But what about $\partial L / \partial \mu$? **It depends.**

If the maximum is found at a point in which $g(x_1, x_2) < 0$:

- **the constraint is slack;**
- the maximum is interior.

Intuition: 2 Variables and 1 Inequality Constraint

When g is slack, the point $\mathbf{x}^*$ is an **unconstrained local maximum** and so

$$\frac{\partial f}{\partial x_1}(\mathbf{x}^*) = \frac{\partial f}{\partial x_2}(\mathbf{x}^*) = 0.$$

If so, we can still use the **Lagrangian**, provided that we set $\mu^* = 0$.

In other words, at a solution $(\mathbf{x}^*, \mu^*)$:

- either the constraint binds, so that: $g(\mathbf{x}^*) = 0$ and $\mu^* \geq 0$;
- or the constraint is slack, so that: $g(\mathbf{x}^*) < 0$ and $\mu^* = 0$.

Definition (Complementary Slackness)

A solution of the constrained optimization problem satisfies the following **complementary slackness** condition:

$$\mu^* g(\mathbf{x}^*) = 0.$$

Intuition: Several Variables

Let f and g be continuously differentiable and let $\mathbf{x}^*$ be the solution to

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad g(\mathbf{x}) \leq 0.$$

Assume also that $\mathbf{x}^*$ is **not a critical point** of $g(\mathbf{x})$ if $g(\mathbf{x}^*) = 0$.

Then given the **Lagrangian function**:

$$L(\mathbf{x}, \mu) = f(\mathbf{x}) - \mu g(\mathbf{x})$$

There **exists a real number** μ^* such that:

$$DL(\mathbf{x}^*) = Df(\mathbf{x}^*) - \mu^* Dg(\mathbf{x}^*) = \mathbf{0},$$

$$\mu^* g(\mathbf{x}^*) = 0,$$

$$\mu^* \geq 0 \quad \text{and} \quad g(\mathbf{x}^*) \leq 0.$$

Inequality Constraints Examples

Example: One Inequality Constraint

Consider a price-taking, profit-maximizing firm.

It produces y from input x according to the production function

$$P(x) = x^{1/2}.$$

The price of output y is 2 and the price of input x is 1.

The firm chooses an amount of units $x \in [0, a]$.

The firm's maximization problem thus satisfies

$$\max_x 2x^{1/2} - x \quad \text{s.t.} \quad x \leq a \quad \text{and} \quad x \geq 0.$$

In the previous notation, $f(x) = 2x^{1/2} - x$ and $g(x) = x - a$.

Example: One Inequality Constraint

Ignoring the constraint $x \geq 0$, the Lagrangian becomes

$$L(x, \mu) = 2x^{1/2} - x - \mu(x - a)$$

The first order condition with respect to x requires that:

$$(x^*)^{-1/2} - 1 - \mu^* = 0$$

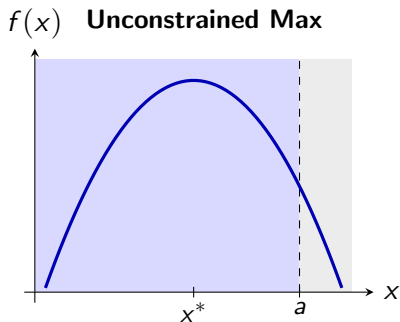
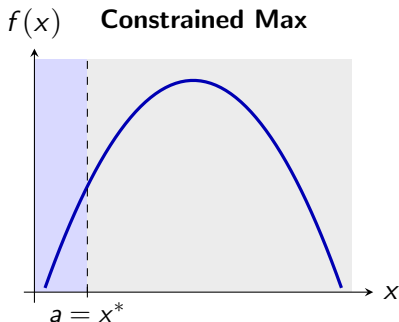
Therefore, a solution to this problem must satisfy:

$$\begin{aligned} (x^*)^{-1/2} - 1 - \mu^* &= 0 \\ \mu^*(x^* - a) &= 0, \quad \mu^* \geq 0 \quad \text{and} \quad x^* \leq a. \end{aligned}$$

To find a maximum, we can now solve the system of equations.

Example: One Inequality Constraint

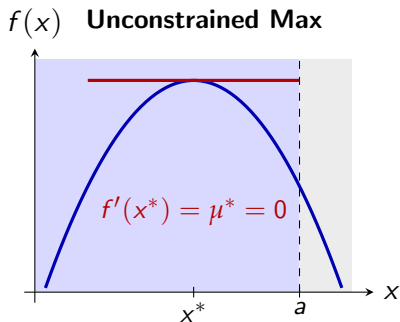
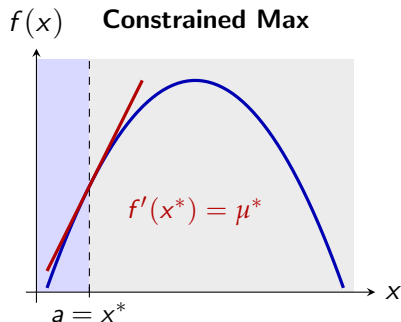
Consider the two separate cases, $\mu^* > 0$ and $\mu^* = 0$:



Remember: $L(x|\mu^*) = f(x) - \mu^*(x - a)$.

Example: One Inequality Constraint

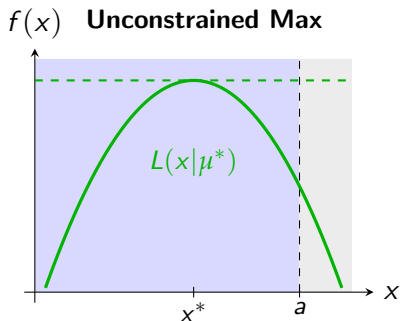
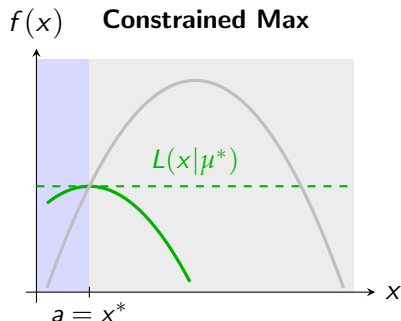
Consider the two separate cases, $\mu^* > 0$ and $\mu^* = 0$:



Remember: $L(x|\mu^*) = f(x) - \mu^*(x - a)$.

Example: One Inequality Constraint

Consider the two separate cases, $\mu^* > 0$ and $\mu^* = 0$:



Remember: $L(x|\mu^*) = f(x) - \mu^*(x - a)$.

Example: One Inequality Constraint

If the **constraint binds**, $x^* = a$ and $\mu^* \geq 0$. If so:

- Necessary conditions require:

$$x^* = a \text{ and } \mu^* = a^{-1/2} - 1.$$

- This solution is viable only if $\mu^* \geq 0$:

$$\mu^* = a^{-1/2} - 1 > 0 \Leftrightarrow a \leq 1.$$

If the **constraint is slack**, $x^* \leq a$ and $\mu^* = 0$. If so:

- Necessary conditions require:

$$x^* = 1 \text{ and } \mu^* = 0.$$

- This solution is viable only if $a \geq 1$.

Example: Several Inequality Constraints

The generalization to several constraints is straightforward.

Key difference:

- some constraints may bind;
- while others may not.

As an example, consider the following problem:

$$\begin{array}{ll}\max_{x_1, x_2, x_3} & x_1 x_2 x_3 \\ \text{s.t.} & x_1 + x_2 + x_3 \leq 1 \\ & x_1 \geq 0, x_2 \geq 0, x_3 \geq 0.\end{array}$$

The Lagrangian for this problem is:

$$L(\mathbf{x}, \boldsymbol{\mu}) = x_1 x_2 x_3 - \mu_0 (x_1 + x_2 + x_3 - 1) + \mu_1 x_1 + \mu_2 x_2 + \mu_3 x_3.$$

Example: Several Inequality Constraints

Solutions of this problem must be a critical point of the Lagrangian.

Also, any solution must satisfy $\mu_1 = \mu_2 = \mu_3 = 0$.

If $\mu_i > 0$, then $x_i = 0$ by complementary slackness.

But this cannot be a maximum since $x_1 x_2 x_3 = 0$, while the objective can be improved by setting

$$x_1 = x_2 = x_3 = 1/10.$$

Thus, **all non-negativity constraints cannot bind**.

This leaves us with a problem with one constraint $x_1 + x_2 + x_3 - 1 \leq 0$.

Thus as before, we only have to decide whether $\mu_0 > 0$ or $\mu_0 = 0$.

Example: Several Inequality Constraints

Obviously, **the constraint binds** and $\mu_0 > 0$.

If the constraint was slack $x_1 + x_2 + x_3 < 1$, we could:

- increase one of the variables,
- satisfy the constraint,
- and increase the value of the function.

Thus, from the first order conditions, we get that

$$x_1 x_2 = x_2 x_3 = x_1 x_3 = \mu_0.$$

FOC coupled with the binding constraint imply that the **solution** satisfies

$$x_1 = x_2 = x_3 = 1/3.$$

Karush-Kuhn-Tucker Theorem

Global Sufficient Conditions

Several Inequality Constraints

Now consider the general problem:

$$\max_{\mathbf{x}} f(\mathbf{x}) \quad \text{s.t.} \quad \mathbf{g}(\mathbf{x}) \leq \mathbf{0},$$

where $\mathbf{g}$ consists of m constraints $\mathbf{g}(\mathbf{x}) = (g_1(\mathbf{x}), \dots, g_m(\mathbf{x}))^T$.

For $\boldsymbol{\mu} = (\mu_1, \dots, \mu_m)$, the Lagrangian of this problem satisfies:

$$L(\mathbf{x}, \boldsymbol{\mu}) = f(\mathbf{x}) - \boldsymbol{\mu} \mathbf{g}(\mathbf{x}).$$

Necessary conditions for a maximum:

$$\begin{aligned} \frac{\partial L(\mathbf{x}^*, \boldsymbol{\mu}^*)}{\partial x_i} &= \frac{\partial f(\mathbf{x}^*)}{\partial x_i} - \boldsymbol{\mu}^* \frac{\partial \mathbf{g}(\mathbf{x}^*)}{\partial x_i} = 0 \quad \text{for any } i \in \{1, \dots, n\}, \\ \mu_j^* \frac{\partial L(\mathbf{x}^*, \boldsymbol{\mu}^*)}{\partial \mu_j} &= 0, \quad \mu_j^* \geq 0 \quad \text{and} \quad g_j(\mathbf{x}^*) \leq 0 \quad \text{for any } j \in \{1, \dots, m\}. \end{aligned}$$

Towards Sufficient Conditions

Such conditions implied that any solution $\mathbf{x}^*$ to a constrained optimization problem would also be a critical point of the Lagrangian.

We have shown how to find the critical points of the Lagrangian.

Question: When can we then say that maximizers of the Lagrangian **also solve the constrained optimization problem?**

Remark

Let $(\mathbf{x}^, \mu^*)$ satisfy all necessary conditions for a maximum.*

If $\mathbf{x}^$ maximizes the Lagrangian, it also maximizes f s.t. $\mathbf{g}(\mathbf{x}) \leq \mathbf{0}$.*

Towards Sufficient Conditions: Proof

Proof of Remark: By complementary slackness,

$$f(\mathbf{x}^*) = f(\mathbf{x}^*) - \mu^* \mathbf{g}(\mathbf{x}^*).$$

Since $\mathbf{x}^*$ maximizes the Lagrangian, then for all other $\mathbf{x}$

$$f(\mathbf{x}^*) - \mu^* \mathbf{g}(\mathbf{x}^*) \geq f(\mathbf{x}) - \mu^* \mathbf{g}(\mathbf{x}).$$

Moreover, by $\mu^* \geq 0$ and $\mathbf{g}(\mathbf{x}) \leq 0$ for all other $\mathbf{x}$ we get that:

$$f(\mathbf{x}) - \mu^* \mathbf{g}(\mathbf{x}) \geq f(\mathbf{x}).$$

But these conditions together imply that $f(\mathbf{x}^*) \geq f(\mathbf{x})$.

So, if $\mathbf{x}^*$ maximizes the Lagrangian, it maximizes $f(\mathbf{x})$ subject to $\mathbf{g}(\mathbf{x}) \leq 0$. $\square$

Towards Sufficient Conditions

In the section on unconstrained optimization we established that:

If L is concave in $\mathbf{x}$ on a convex subset $U \subset \mathbb{R}^n$ and $\mathbf{x}^*$ a point in the interior of U such that $DL(\mathbf{x}^*, \boldsymbol{\mu}) = 0$, then $\mathbf{x}^*$ maximizes $L(\mathbf{x}, \boldsymbol{\mu})$ in U :

$$L(\mathbf{x}, \boldsymbol{\mu}) \leq L(\mathbf{x}^*, \boldsymbol{\mu}) \quad \text{for all } \mathbf{x} \in U.$$

Moreover, because any sum of concave functions is concave::

The Lagrangian is concave whenever f is concave and g is convex.

Thus, in such environments:

First order conditions must be both necessary and sufficient.

Karush-Kuhn-Tucker Theorem: Inequality Constraints

Definition (KKT Constraint Qualification - Slater Condition)

There exists $\mathbf{x} \in \mathbb{R}^n$ such that $\mathbf{g}(\mathbf{x}) < \mathbf{0}$.

Theorem (Karush-Kuhn-Tucker Theorem)

Let f and $\mathbf{g}$ be differentiable. Let f be concave, g_j be convex for all $j \in \{1, \dots, m\}$, and the constraint qualification hold. Then $\mathbf{x}^$ solves the constraint optimization problem if and only if $(\mathbf{x}^*, \boldsymbol{\mu}^*)$ solves*

$$\begin{aligned} Df(\mathbf{x}^*) - \boldsymbol{\mu}^* D\mathbf{g}(\mathbf{x}^*) &= \mathbf{0}, \\ \boldsymbol{\mu}^* \mathbf{g}(\mathbf{x}^*) &= \mathbf{0}, \quad \boldsymbol{\mu}^* \geq \mathbf{0} \quad \text{and} \quad \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}. \end{aligned}$$

The theorem delivers global second order conditions.

Karush-Kuhn-Tucker Theorem: All Constraints

Definition (KKT Constraint Qualification)

The gradients of the binding inequality constraints and the gradients of the equality constraints are linearly independent at the solution $\mathbf{x}^*$.

Theorem (Karush-Kuhn-Tucker Theorem)

Let f , $\mathbf{h}$ and $\mathbf{g}$ be differentiable. Let f be concave, g_j be convex for all $j \in \{1, \dots, m\}$, h_j be convex for all $j \in \{1, \dots, k\}$, and the constraint qualification hold. Then $\mathbf{x}^$ solves the constraint optimization problem if and only if $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\mu}^*)$ solves*

$$\begin{aligned} Df(\mathbf{x}^*) - \boldsymbol{\lambda}^* D\mathbf{h}(\mathbf{x}^*) - \boldsymbol{\mu}^* D\mathbf{g}(\mathbf{x}^*) &= \mathbf{0}, \\ \mathbf{h}(\mathbf{x}^*) &= \mathbf{0}, \quad \boldsymbol{\mu}^* \mathbf{g}(\mathbf{x}^*) = \mathbf{0}, \quad \boldsymbol{\mu}^* \geq \mathbf{0} \quad \text{and} \quad \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0}. \end{aligned}$$

Arrow-Enthoven Theorem

The KKT Theorem applies to concave objective functions and convex constraints. Many economic objective functions are however only quasi-concave, and many constraints are only quasi-convex.

Theorem (Arrow-Enthoven Theorem)

Let f and $\mathbf{g}$ be differentiable. If f is quasi-concave and g_j is quasi-convex for all j , then any solution $(\mathbf{x}^, \boldsymbol{\mu}^*)$ to the KKT conditions,*

$$D_{\mathbf{x}}L(\mathbf{x}^*, \boldsymbol{\mu}^*) = \mathbf{0}, \quad \boldsymbol{\mu}^* D_{\boldsymbol{\mu}}L(\mathbf{x}^*, \boldsymbol{\mu}^*)^T = 0, \quad \boldsymbol{\mu}^* \geq \mathbf{0} \quad \text{and} \quad \mathbf{g}(\mathbf{x}^*) \leq \mathbf{0},$$

solves the constraint optimization problem provided that $D_{\mathbf{x}}f(\mathbf{x}^) \neq \mathbf{0}$ and that $f(\mathbf{x})$ is twice differentiable in the neighborhood of $\mathbf{x}^*$.*

Alternative regularity conditions can replace the latter requirement.

Adding equality constraints is no problem provided these are quasi-convex.

Extra: Local Sufficient Conditions

Extra: Local Sufficient Conditions

When global sufficient conditions do not apply consider a point $(\mathbf{x}^*, \mu^*)$ which solves the KKT necessary conditions.

Assume that g_1 to g_e are binding and that g_{e+1} to g_m are slack so that:

$$\mu_{e+1} = 0, \dots, \mu_m = 0.$$

Conveniently define:

$$\boldsymbol{\mu}_E = (\mu_1, \dots, \mu_e) \quad \text{and} \quad \mathbf{g}_E = (g_1, \dots, g_e)^T.$$

Let the Hessian of L with respect to $\mathbf{x}$ at $(\mathbf{x}^*, \mu^*)$ be **negative definite on the linear constraint set** $\{\mathbf{v} \mid D_{\mathbf{x}}\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = \mathbf{0}\}$ so that:

$$\mathbf{v} \neq \mathbf{0} \quad \text{and} \quad D_{\mathbf{x}}\mathbf{g}_E(\mathbf{x}^*)\mathbf{v} = \mathbf{0} \quad \implies \quad \mathbf{v}^T (D_{\mathbf{x}}^2 L(\mathbf{x}^*, \mu^*)) \mathbf{v} < 0.$$

If so, $\mathbf{x}^*$ is a **strict local maximizer of f on the constraint set.**

Question:

How can we show that $D_{\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\mu}^*)$ is negative definite on constraint set?

To establish this we make use of the **Bordered Hessian**:

$$B = \begin{pmatrix} \mathbf{0} & -D_{\mathbf{x}} g_E(\mathbf{x}^*) \\ -D_{\mathbf{x}} g_E(\mathbf{x}^*)^T & D_{\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\mu}^*) \end{pmatrix}$$

If the last $(n - e)$ leading principal minors of B alternate in sign with the sign of the determinant of the largest matrix the same as the sign of $(-1)^n$, then **local sufficient second order conditions** hold for a candidate point $\mathbf{x}^*$ to be a local solution to a constrained maximization problem.

Extra: Local SOC Intuition

The bordered Hessian is the Hessian of Lagrangian $D^2L(\boldsymbol{\mu}, \mathbf{x})$ obtained by eliminating slack constraints.

$$\begin{pmatrix} D_{\boldsymbol{\mu}}^2 L(\boldsymbol{\mu}, \mathbf{x}) = \mathbf{0} & \frac{\partial^2 L}{\partial \mu_1 \partial x_1} = -\frac{\partial g_1}{\partial x_1} & \cdots & \frac{\partial^2 L}{\partial \mu_1 \partial x_n} = -\frac{\partial g_1}{\partial x_n} \\ & \vdots & \ddots & \vdots \\ & \frac{\partial^2 L}{\partial \mu_e \partial x_1} = -\frac{\partial g_e}{\partial x_1} & \cdots & \frac{\partial^2 L}{\partial \mu_e \partial x_n} = -\frac{\partial g_e}{\partial x_n} \\ D_{\boldsymbol{\mu}, \mathbf{x}}^2 L(\boldsymbol{\mu}, \mathbf{x})^T & \frac{\partial^2 L}{\partial x_1^2} & \cdots & \frac{\partial^2 L}{\partial x_1 \partial x_n} \\ & \vdots & \ddots & \vdots \\ & \frac{\partial^2 L}{\partial x_1 \partial x_n} & \cdots & \frac{\partial^2 L}{\partial x_n^2} \end{pmatrix}$$

Extra: Local SOC Example I

Consider the problem:

$$\max_{x_1, x_2} f(x_1, x_2) \quad \text{s.t.} \quad g_1(x_1, x_2) \leq 0.$$

Assume that there exist a solution to the KKT conditions:

$$\begin{aligned} D_x L(x_1^*, x_2^*, \mu^*) &= \mathbf{0}, \\ \mu^* &\geq 0, \quad g_1(x_1^*, x_2^*) \leq 0 \quad \text{and} \quad \mu^* g_1(x_1^*, x_2^*) = 0. \end{aligned}$$

If $\mu^* > 0$, the bordered Hessian of this problem is:

$$B = \begin{pmatrix} 0 & -\frac{\partial g}{\partial x_1} & -\frac{\partial g}{\partial x_2} \\ -\frac{\partial g}{\partial x_1} & \frac{\partial^2 f}{\partial x_1^2} - \mu^* \frac{\partial^2 g}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} - \mu^* \frac{\partial^2 g}{\partial x_1 \partial x_2} \\ -\frac{\partial g}{\partial x_2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} - \mu^* \frac{\partial^2 g}{\partial x_1 \partial x_2} & \frac{\partial^2 f}{\partial x_2^2} - \mu^* \frac{\partial^2 g}{\partial x_2^2} \end{pmatrix}.$$

If so, the local SOC require: $|B| > 0$ at (x_1^*, x_2^*) .

Extra: Local SOC Example II

Consider now the problem:

$$\max_{x_1, x_2, x_3} f(x_1, x_2, x_3) \quad \text{s.t.} \quad g(x_1, x_2, x_3) \leq 0.$$

The bordered Hessian matrix is:

$$B = \begin{pmatrix} 0 & -\frac{\partial g}{\partial x_1} & -\frac{\partial g}{\partial x_2} & -\frac{\partial g}{\partial x_3} \\ -\frac{\partial g}{\partial x_1} & \frac{\partial^2 L}{\partial x_1^2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_3} \\ -\frac{\partial g}{\partial x_2} & \frac{\partial^2 L}{\partial x_1 \partial x_2} & \frac{\partial^2 L}{\partial x_2^2} & \frac{\partial^2 L}{\partial x_2 \partial x_3} \\ -\frac{\partial g}{\partial x_3} & \frac{\partial^2 L}{\partial x_1 \partial x_3} & \frac{\partial^2 L}{\partial x_2 \partial x_3} & \frac{\partial^2 L}{\partial x_3^2} \end{pmatrix}$$

Local SOC require: $|B_3| > 0$ and $|B_4| < 0$.